

CS333 Midterm 1 Part (a)

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I. MULTIPLE CHOICE QUESTIONS (50 POINTS)

1. What is the value of *sum* after the following code fragment is executed?

```
int sum = 0;
for (i = 1; i <= N; i++)
    for (j = 2 * i; j <= 2 * N; j += 2)
        sum++;
```

- a) N
- b) $(N^2 + N)/2$
- c) N^2
- d) N^3
- e) N^4

2. Given the time complexity equation for a recursive algorithm as $T(N) = 9T(N/3) + \mathcal{O}(N^2)$. What is the time complexity of that algorithm?

- a) $\mathcal{O}(N)$
- b) $\mathcal{O}(N \log N)$
- c) $\mathcal{O}(N^2)$
- d) $\mathcal{O}(N^2 \log N)$
- e) $\mathcal{O}(N^3)$

3. What is the maximum number of edges in a directed graph without loops for N nodes?

- a) N
- b) N^2
- c) $(N^2 + N)/2$
- d) $(N^2 - N)/2$
- e) N^3

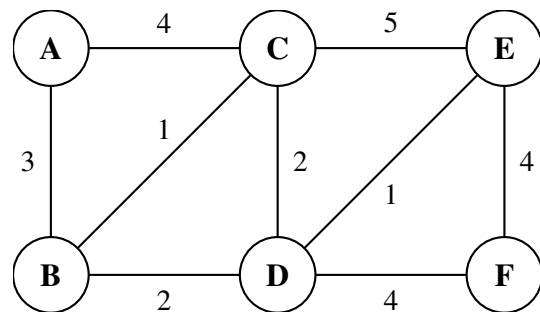
4. What is the worst case time complexity of the algorithm (for a graph with N nodes) which checks if node i has no incoming edges (i) adjacency matrix, (ii) adjacency list representation?

- a) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
- b) $\mathcal{O}(\log N)$, $\mathcal{O}(1)$
- c) $\mathcal{O}(N)$, $\mathcal{O}(1)$
- d) $\mathcal{O}(N)$, $\mathcal{O}(N^2)$
- e) $\mathcal{O}(1)$, $\mathcal{O}(N)$

5. If you want to solve the single source multiple destination shortest path problem, which algorithm should you use?

- a) Dijkstra
- b) Floyd-Warshall
- c) Bellman-Ford
- d) Kruskal
- e) Prim

6. Given the graph below, what is the length of the



minimum spanning tree?

- a) 9
- b) 10
- c) 11
- d) 12
- e) 13

7. What is the length of the minimum spanning tree of a graph with N nodes with equal edge weights of 1?

- a) N^2
- b) N
- c) $N + 1$
- d) $N - 1$
- e) $N / 2$

8. What is the minimum number of connected components of a graph with N nodes?

- a) $N / 4$
- b) N
- c) 2
- d) $N - 1$
- e) 1

9. Given the graph in Question 6, which of the following is a possible list of four selected edges in Kruskal's Minimum Spanning Tree algorithm?

- a) BC, BD, CD, DE
- b) BC, DE, EF, DF
- c) AB, BC, DE, BD
- d) AB, BC, BD, CD
- e) AC, DF, EF, CE

10. Given the graph in Question 6, which of the following is a possible list of four selected edges in Prim's Minimum Spanning Tree algorithm, if the algorithms start from node A?

- a) AB, AC, CD, BD
- b) AB, BC, CE, BD
- c) AB, BC, CD, DE
- d) AC, BC, CD, DE
- e) AC, CD, BC, CE

11. Which of the following functions has the largest growth?

- a) n^2
- b) n
- c) $n \log(n)$
- d) 1.01^n
- e) n^{10}

12. Given the time complexity equation for a recursive algorithm as $T(N) = 8T(N/2) + \mathcal{O}(N^2)$. What is the time complexity of that algorithm?

- a) $\mathcal{O}(N^2 \log N)$
- b) $\mathcal{O}(N^2)$
- c) $\mathcal{O}(N^3)$
- d) $\mathcal{O}(N \log N)$
- e) $\mathcal{O}(N)$

13. What is the worst case time complexity of the algorithm (for a graph with N nodes) which checks if node i is connected to node j with (i) adjacency matrix, (ii) adjacency list representation?

- a) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
- b) $\mathcal{O}(\log N)$, $\mathcal{O}(1)$
- c) $\mathcal{O}(N)$, $\mathcal{O}(1)$
- d) $\mathcal{O}(1)$, $\mathcal{O}(1)$
- e) $\mathcal{O}(1)$, $\mathcal{O}(N)$

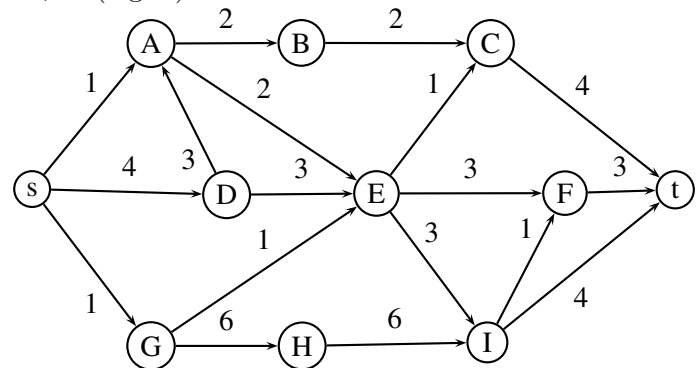
14. For an unweighted undirected graph G with N nodes and E edges, what is the sum of the elements of the adjacency matrix of G ?

- a) E
- b) $2E$
- c) N
- d) $2N$
- e) N^2

15. What is the time complexity of the following fragment of code?

```
int i = 1;
while (i < N * N) {
    i = i * 2;
}
```

- a) $\mathcal{O}(N)$
- b) $\mathcal{O}(1)$
- c) $\mathcal{O}(N^2)$
- d) $\mathcal{O}(\sqrt{N})$
- e) $\mathcal{O}(\log N)$



16. In the graph above, what is the shortest path from s to t ?

- a) 11
- b) 10
- c) 9
- d) 8
- e) 7

17. Given the graph in Question 16, which of the following is a possible list of four selected nodes in Dijkstra's shortest path algorithm, if the algorithms start from node s ?

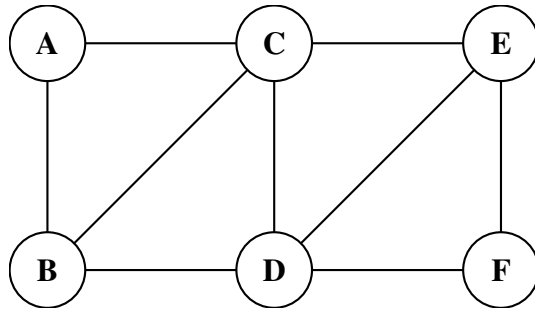
- a) A, D, G, E
- b) A, G, H, B
- c) A, D, G, E
- d) A, G, E, B
- e) A, G, D, E

18. Which problem can be easily solved by using breath first search?

- a) Single source multiple destination shortest path problem in weighted graph
- b) Multiple source multiple destination shortest path problem in weighted graph
- c) Single source multiple destination shortest path problem in unweighted graph
- d) Multiple source multiple destination shortest path problem in negative weighted graph
- e) None

19. Which of the following denominations can not be solved optimally by the greedy algorithm (Coin Change problem)?

- a) 1, 5, 10, 20
- b) 1, 10, 20, 100
- c) 1, 5, 8, 20
- d) 1, 4, 8, 16
- e) All of them can be solved



20. Given the graph above, which of the following is a possible list for the BFS traversal of the graph starting from node A?

- a) ABCDEF
- b) ABDFEC
- c) ABCDFE
- d) ACEFDB
- e) ACDBEF

21. Which of the following greedy strategy is optimal for interval partitioning problem? In interval partitioning problem, the goal is to find the minimum number of classrooms to schedule so that no two occur at the same time in the same room. In the end, all lectures must be scheduled.

- a) Lecture having the earliest start time is selected first
- b) Lecture having the earliest finish time is selected first
- c) Lecture having fewest conflict is selected first
- d) None of them are optimal
- e) All of them are optimal

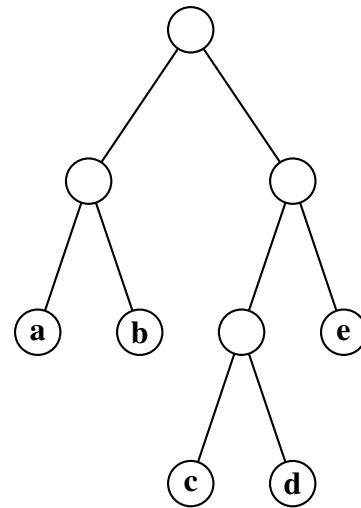
22. Given the graph in Question 20, which of the following is a possible list for the DFS traversal of the graph starting from node A?

- a) ACEFBD
- b) ABCEFD
- c) ABDFCE
- d) ABDCFE
- e) ACEBDF

23. Which of the following greedy strategy is optimal for interval scheduling problem? In interval scheduling problem, the maximum subset of mutually compatible jobs is determined. In the end, some jobs may not be scheduled.

- a) Job having the earliest start time is selected first
- b) Job having the earliest finish time is selected first
- c) Job having the shortest interval is selected first
- d) Job having fewest conflict is selected first
- e) All of them are optimal

24. Given the Huffman tree above, what is the com-



pressed form of the string “abdcde”? Assume that, left children are represented by 1 and right children are represented by 0.

- a) 0001101101000
- b) 100110100010101
- c) 011011001001
- d) 111001001101000
- e) 11010111001

25. What is the minimum number of bits used to represent the string “abcde” according to Huffman encoding, if the number of occurrences of the characters are:

a = 1, b = 2, c = 4, d = 8, e = 16.

- a) 10
- b) 12
- c) 14
- d) 16
- e) 18